

Nilesh Mishra

+91 8652479323 | mishranilesh968@gmail.com | LinkedIn | Portfolio | GitHub

SUMMARY

AI Engineer specializing in **Computer Vision, Generative AI (LLMs, RAG), and Time-Series Forecasting**. Experienced in building **production-grade ML systems** using FastAPI, LangChain, and deep learning frameworks. Delivered scalable AI solutions across **healthcare, automation, and analytics**, improving efficiency and decision-making.

TECHNICAL SKILLS

Languages: Python, C/C++, SQL

Frameworks: FastAPI, Flask, LangChain, TensorFlow, PyTorch, Keras, Scikit-learn, Pydantic

Tools: OpenCV, Hugging Face, Pinecone, Tesseract, Docker, Git, AWS

Domains: Computer Vision, NLP, RAG, OCR Systems, Time-Series Forecasting, MLOps

EXPERIENCE

AI Engineer Intern

Nov 2025 – Present

Bhatia Brothers FZE

- Built **inventory analytics platform** to track stock movement and automate reorder cycles, reducing manual workload by **30–40%**.
- Developed **time-series forecasting models** (Prophet-based) improving demand prediction accuracy by **20%+**.
- Designed **OCR + NLP pipeline** to process invoices/reports, automating **1000+ documents/month** with high accuracy.
- Deployed scalable APIs using **FastAPI**, enabling real-time inventory insights and faster decision-making.

Junior Software Developer

May 2025 – Sept 2025

Enirmaan Techsolutions Pvt. Ltd.

- Engineered a **real-time health vitals monitoring system** using OpenCV + FastAPI, enabling contactless patient tracking.
- Improved system accuracy by **25%** via face validation and multi-face rejection algorithms.
- Built **OCR + LLM pipeline** extracting structured data from images with automated validation.

Tech Support Intern

Oct 2024 – Feb 2025

Livingthings.ai (IIT Bombay)

- Analyzed **IoT sensor data** to optimize AC energy usage, contributing to measurable energy efficiency improvements.

Desktop Support Intern

Mar 2024 – Jun 2024

Policy Junction Marketing Pvt. Ltd.

- Reduced system downtime by **20%** through efficient troubleshooting and system optimization.

PROJECTS

Health Vitals Monitoring via Video

FastAPI, OpenCV

- Developed a **contactless vitals estimation system** enabling real-time healthcare monitoring with computer vision.

Medical Chatbot with RAG

LangChain, Pinecone, OpenAI

- Built a **context-aware medical assistant** using retrieval-augmented generation for accurate Q&A.

OCR-Based Visiting Card Extractor

Tesseract, LLMs

- Designed a **hybrid OCR + LLM pipeline** for structured data extraction with validation.

Stock Price Forecasting

LSTM, CNN-LSTM

- Implemented deep learning models for **time-series prediction** with feature engineering.

EDUCATION

M.Sc. in Information Technology

2023 – 2025

S.M. Shetty College, Mumbai

CGPA: 7.2/10

B.Sc. in Information Technology

2020 – 2023

S.M. Shetty College, Mumbai

CGPA: 8.18/10

ADDITIONAL

Certifications: Machine Learning, EDA, AWS ML Basics

Interests: Generative AI, Computer Vision, NLP, AI in Healthcare